SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	O. Siva Charan	**Reg. No.:**	192525039
Section / Batch:	AI&ML/2025	**Date:**	24/07/2026

Code of Conduct

I O. Siva Charan (192525039)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Siva Charan__________

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks

Q1 **Binary Search** 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:/CCPPComp/CCPP3/bin/../lib/gcc/x8
```

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

Q2 Stack 5 marks

Find the error in the stack deletion operation. Justify it.

int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int stack[MAX];
int top = -1;
void push(int value) {
    if (top == MAX - 1) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = value;
}
int pop() {
    if (top == -1) {
```

Test Input

stdin input (optional)

Output

```
Popped: 30
Popped: 20
Popped: 10
Underflow
Popped: -1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1 Binary Search 5 marks
Q2 Stack 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Stack - Balancing Symbols 5 marks
Q6 Singly Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List 5 marks
Q9 Linked List 5 marks

Q3 **Stack**

5 marks

Check the snippet and identify the logical error.
int stack[5];
int top;
void init() {
top = 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define SIZE 5

int stack[SIZE];
int top;

void init() {
    top = -1;
}

int isEmpty() {
    return top == -1;
}

int isFull() {
    return top == SIZE - 1;
}
```

Test Input

stdin input (optional)

Output

```
Top element: 30
Popped: 30
Top element after pop: 20
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q4 **Stack** 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != '(') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX 100
struct Stack {
    int top;
    char items[MAX];
};
void push(struct Stack* s, char c) {
    if (s->top < MAX - 1) {
        s->items[++(s->top)] = c;
    }
}

char pop(struct Stack* s) {
```

Test Input

```
a+b*c
```

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q5 **Stack - Balancing Symbols** 5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 100
int stack[MAX];
int top = -1;
void push(int val) {
    if (top == MAX - 1) {
        return;
    }
    stack[++top] = val;
}
int pop() {
    if (top == -1) {
        return -1;
    }
```

Test Input

```
{[()]}
```

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks

Q6 **Singly Linked List** 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
void printList(struct Node* head) {
    struct Node* temp = head;
    while (temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}
struct Node* newNode(int data) {
    struct Node* node = (struct Node*)malloc(sizeof(struct Node));
    node->data = data;
    node->next = NULL;
    return node;
}
```

Test Input

stdin input (optional)

Output

```
1
2
3
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks

Q7 **Stack** 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

int main() {
    int stack[MAX];
    int top = -1;
    stack[++top] = 10;
    stack[++top] = 20;
    if (top == -1) {
        printf("Underflow\n");
    } else {
        int x = stack[top];
        top = top - 1;
        printf("Popped: %d\n", x);
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Popped: 20
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks

Q8 **Linked List** 5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* reverseList(struct Node* head) {
    struct Node* prev = NULL;
    struct Node* current = head;
    struct Node* next_node = NULL;

    while (current != NULL) {
        next_node = current->next;
        current->next = prev;
        prev = current;
        current = next_node;
    }
    return prev;
}
void printList(struct Node* head) {
```

Test Input

stdin input (optional)

Output

```
Original Linked List:
1 -> 2 -> 3 -> 4 -> NULL
Reversed Linked List:
4 -> 3 -> 2 -> 1 -> NULL
```

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10

Q9 **Linked List**

5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

typedef struct Node {
    int data;
    struct Node *next;
} Node;

void insertAtBeginning(Node **head, int value) {
    Node *newNode = (Node *)malloc(sizeof(Node));
    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        exit(1);
    }

    newNode->data = value;
    newNode->next = *head;
    *head = newNode;
}

void printList(Node *head) {
    while (head != NULL) {
        printf("%d ", head->data);
        head = head->next;
    }
```

Test Input

stdin input (optional)

Output

```
30 20 10
```

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10

Q10 **Queue**

5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 5

struct Queue {
    int items[MAX];
    int front;
    int rear;
};
void initialize(struct Queue *q) {
    q->front = -1;
    q->rear = -1;
}
int isFull(struct Queue *q) {
    if (q->rear == MAX - 1) {
        return 1;
    }
    return 0;
}
int isEmpty(struct Queue *q) {
```

Test Input

stdin input (optional)

Output

```
Inserted: 10
Inserted: 20
Inserted: 30
Queue elements: 10 20 30
Dequeued: 10
Dequeued: 20
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks

Q1 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 100
int stack[MAX];
int top = -1;
void push(int value) {
    if (top >= MAX - 1) {
        printf("Stack Overflow\n");
        return;
    }
    stack[++top] = value;
}
int PEEK() {
    if (top == -1) {
        printf("Stack is empty\n");
        return -1;
    }
    return stack[top];
}

int main() {
    push(10);
    push(20);
```

Test Input

```
stdin input (optional)
```

Output

```
Top element is: 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue

Q2 **Circular Queue** 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5
int queue[MAX];
int front = 0;
int rear = -1;
void enqueue(int vehicle) {
    int next = (rear + 1) % MAX;
    if (next == front) {
        printf("Queue Full\n");
    } else {
        rear = next;
        queue[rear] = vehicle;
        printf("Vehicle %d added\n", vehicle);
    }
}
int main() {
    enqueue(10);
    enqueue(20);
```

Test Input

stdin input (optional)

Output

```
Queue Full
Queue Full
Queue Full
Queue Full
Queue Full
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks

Q3 **Stack** 5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Overflow\n");
    } else {
        top++;
        stack[top] = value;
    }
}

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    } else {
```

Test Input

```
6
1
10
1
```

Output

```
Enter number of operations: Enter
choice (1=push, 2=pop, 3=display):
Enter element: Enter choice
(1=push, 2=pop, 3=display): Enter
element: Enter choice (1=push,
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks

Q4 **Stack** 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

typedef struct {
    int arr[MAX];
    int top;
} Stack;
void initStack(Stack *s) {
    s->top = -1;
}
int isEmpty(Stack *s) {
    return s->top == -1;
}
int isFull(Stack *s) {
    return s->top == MAX - 1;
}
void push(Stack *s, int x) {
    if (isFull(s)) {
        printf("Overflow\n");
        return;
```

Test Input

stdin input (optional)

Output

```
Popped: 20
Popped: 10
Underflow
Popped: -1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue

Q5 **Doubly Linked List** 5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* prev;
    struct Node* next;
};

typedef struct Node Node;
Node* createNode(int data) {
    Node* newNode = (Node*)malloc(sizeof(Node));
    newNode->data = data;
    newNode->prev = NULL;
    newNode->next = NULL;
    return newNode;
}
void insertEnd(Node** head, int data) {
```

Test Input

stdin input (optional)

Output

```
Original list: 10 20 30 40
List after deletion: 10 20 40
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int data) {
    if (rear == MAX - 1) {
        printf("Queue Overflow\n");
        return;
    }
    if (front == -1) {
        front = 0;
    }
    queue[++rear] = data;
    printf("%d enqueued to queue\n", data);
}

```

Test Input

stdin input (optional)

Output

```
10 enqueued to queue
20 enqueued to queue
30 enqueued to queue
Queue elements: 10 20 30
Dequeued: 10
Queue elements: 20 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks

Q7 **Queue - Deque** 5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5

int deque[MAX];
int front = -1;
int rear = -1;
int isFull() {
    return ((front == 0 && rear == MAX - 1) || (front == rear + 1));
}
int isEmpty() {
    return (front == -1);
}
void insertFront(int data) {
    if (isFull()) {
        printf("Overflow: Deque is full\n");
        return;
    }

    if (front == -1) {
```

Test Input

stdin input (optional)

Output

```
Inserted 10 at rear
Inserted 20 at rear
Inserted 5 at front
Inserted 1 at front
Deque elements: 1 5 10 20
Deleted 1 from front
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue

Q8 **Binary Search** 5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x) {
    if (r < l) return -1;

    int mid = l + (r - l) / 2;

    if (arr[mid] == x) return mid;

    if (arr[mid] > x)
        return binarySearch(arr, l, mid - 1, x);
    else
        return binarySearch(arr, mid + 1, r, x);
}

int main() {
    int n;
```

Test Input

```
10
```

Output

```
-1
```

Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;
void enqueue(int value) {
    if ((rear + 1) % MAX == front) {
        printf("Queue is Full\n");
        return;
    }
    if (front == -1) {
        front = 0;
    }
    rear = (rear + 1) % MAX;
    queue[rear] = value;
```

Test Input

stdin input (optional)

Output

```
Inserted: 10
Inserted: 20
Inserted: 30
Inserted: 40
Inserted: 50
Queue elements: 10 20 30 40 50
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10 **Stack - Postfix Expression** 5 marks

Find the bug in the evaluation of a Postfix expression.
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

#define MAX 100

int stackArr[MAX];
int top = -1;

void push(int x) {
    if (top >= MAX - 1) return;
    stackArr[++top] = x;
}

```

Test Input

stdin input (optional)

Output

0